

UNIT 1

Introduction

Sessions 1 and 2

Every machine in this book has to answer one question well: *how far?* Measuring accurately turns out to be harder, and far more interesting, than it sounds. This unit gives you the electrical foundation you will need, and then everything after it is building.

By the end of this unit you will be able to:

- Explain the Sense - Think - Act idea with your own example.
- Name every component in your kit and say what it does.
- Build a circuit on a breadboard and explain why it works.
- Tell a digital signal from an analog one, and say which pins handle each.

SESSION 1

one class

Machines That Measure, and What Is in Your Box

Today you will

- Understand why measurement is the heart of robotics.
- Learn the names of all twelve components in your kit.
- Agree the rules of the robotics lab.

A robot is any machine that can **sense** something about the world, **decide** what that means, and then **act** on the decision. The automatic door at a shop, the washing machine that slows down when the drum is unbalanced, and the reversing camera in a car are all robots by that definition, even though none of them looks like one.

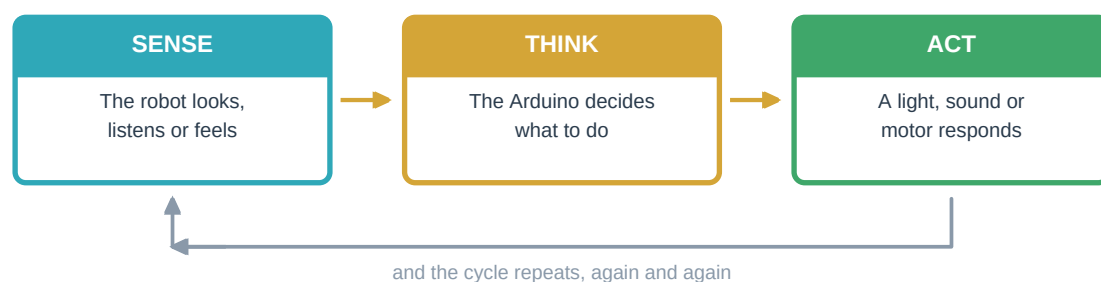


Figure 1.1 - Every robot in the world follows this same three-step loop

This year the **sense** step gets serious. Deciding whether a room is dark is easy - you only need to be roughly right. But if your robot thinks a wall is 40 centimetres away when it is really 20, it will drive straight into it. Once a machine has to move, accuracy stops being a nice extra and becomes the whole job.

Try This

Guess the width of your desk in centimetres. Write your guess down. Now measure it with a ruler and write the real value beside it. The difference between the two numbers has a name that you will use all year: it is called **error**. Engineers do not try to have zero error. They try to know how big their error is.

1.1 Open the Box

Lay everything on the desk before you plug anything in. Learn the names first - it is impossible to ask for help with 'that small blue thing'. Four of these will be completely new to most of you: the ultrasonic sensor, the servo motor, the motor driver and the chassis.

ARDUINO KIT COMPONENTS – CLASS 7

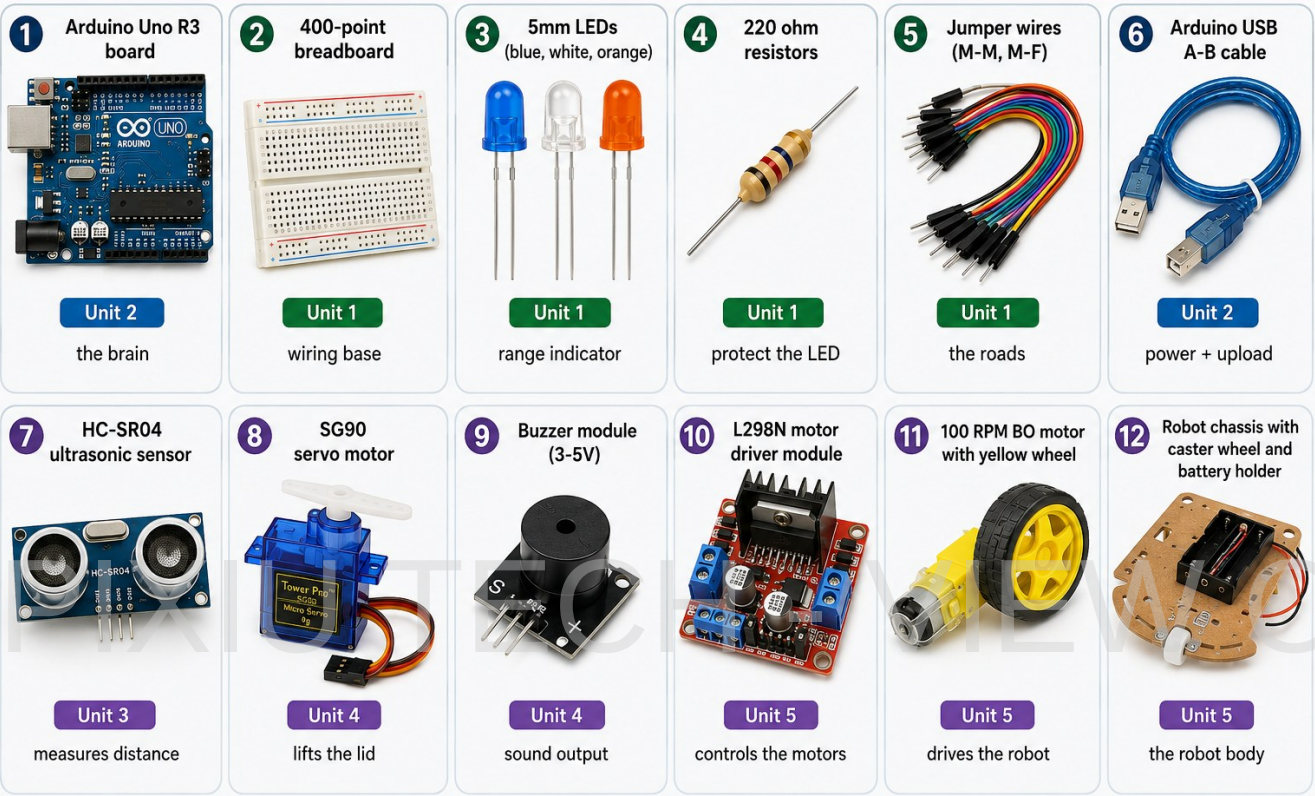


Figure 1.1 - Your kit. Learn these twelve names before the next class.

Part	What it does	First used in
Arduino Uno board	The brain. It stores your program and runs it forever, even after the power has been off.	Session 3
USB A-B cable	Carries your program into the board and powers it while you test.	Session 3
400-point breadboard	Joins components together without any soldering.	Session 2
Jumper wires	The roads between parts. Male-to-male and male-to-female.	Session 2
5mm LEDs	Small lamps. Here they become a distance indicator you can read at a glance.	Session 2
220 ohm resistors	Limit the current so an LED is not destroyed.	Session 2
HC-SR04 ultrasonic sensor	Measures distance by timing an echo. The star of this book.	Session 6
SG90 servo motor	Turns to an exact angle and holds it there.	Session 9
Buzzer module	Makes sound, so the machine can warn people.	Session 10
L298N motor driver	A power amplifier for motors. The Arduino cannot drive them directly.	Session 12
100 RPM BO motor with wheel	Turns the wheels. You need two of these.	Session 13
Chassis, caster wheel, battery holder	The robot's body and its power supply.	Session 13

1.2 Rules of the Robotics Lab

These eight rules are not decoration. Each one exists because somebody, somewhere, damaged equipment by ignoring it. Two of them are new this year, because this year you will work with batteries and motors.

- 1 Always unplug the USB cable before you change any wire.
- 2 Check the long leg of every LED and the + side of every module before power goes on.
- 3 Never run a wire straight from 5V to GND. That is a short circuit and it can kill the board.
- 4 **Never connect a motor directly to an Arduino pin.** It draws far more current than the pin can supply and will damage the board permanently.
- 5 **Disconnect the battery pack before rewiring anything**, and never leave it connected at the end of a session.
- 6 Lift a robot off the floor before uploading new code, so it cannot drive away mid-upload.
- 7 If anything smells hot or feels hot, disconnect it at once and call your trainer.
- 8 Count every component back into the kit box before you leave.

Watch Out

Rule 4 is the one that ends most school robotics kits. An Arduino pin can supply about 20 milliamps safely. A BO motor wants several hundred. That gap is exactly why the L298N motor driver exists, and you will meet it properly in Unit 5.

New word	What it means
Robot	A machine that senses, decides and acts.
Error	The difference between a measured value and the true value.
Current	How much electricity flows each second. Measured in amps.

SESSION 2

one class

Circuits, Breadboards and Signals

Today you will

- Refresh how a circuit works and why it must be a loop.
- Use a breadboard correctly and light an LED with no code.
- Learn the difference between digital and analog signals.

Electricity behaves much like water in a pipe. The battery is the pump, the wires are the pipes, and the LED is a small fountain at the end. **Voltage** is how hard the pump pushes, measured in volts. **Current** is how much flows each second, measured in amps. **Resistance** is anything that makes flowing harder, measured in ohms.

The rule that matters most: **electricity only flows in a complete loop**. It must leave the supply, pass through every component, and return. Break the loop anywhere and everything stops. A switch is simply a gap you can open and close deliberately.

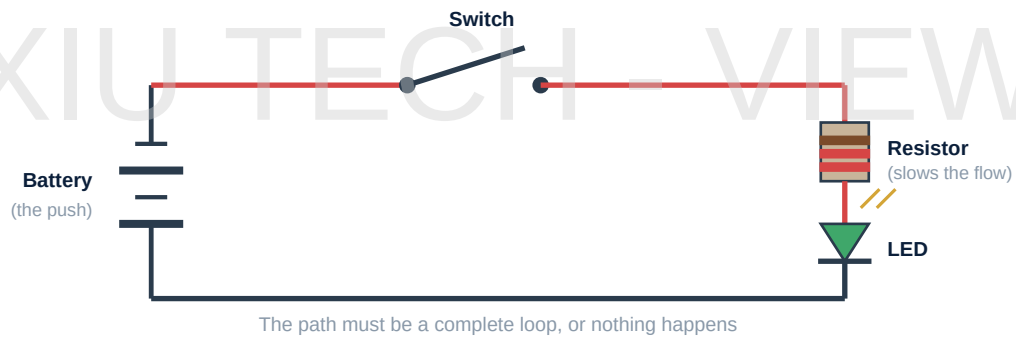
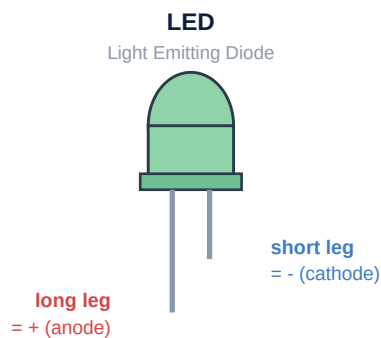


Figure 1.2 - A simple circuit: battery, switch, resistor and LED



Put the long leg towards +. Backwards, it simply will not glow (and that is not a fault).



Its job is to protect the LED from too much current, like a narrow gate in a wide road.

Watch Out

An LED has a long leg and a short leg. Long leg towards +, short leg towards -. If an LED does not glow, turn it around before checking anything else. And it always needs a resistor beside it: connect one straight to 5V and it will glow brilliantly for about a second, then never again.

2.1 Inside the Breadboard

A breadboard looks like a plain block of holes but is full of hidden metal strips. Two holes joined by a strip behave exactly as though you had twisted those wires together. Knowing which holes are secretly joined is the commonest reason a beginner's circuit does nothing at all.

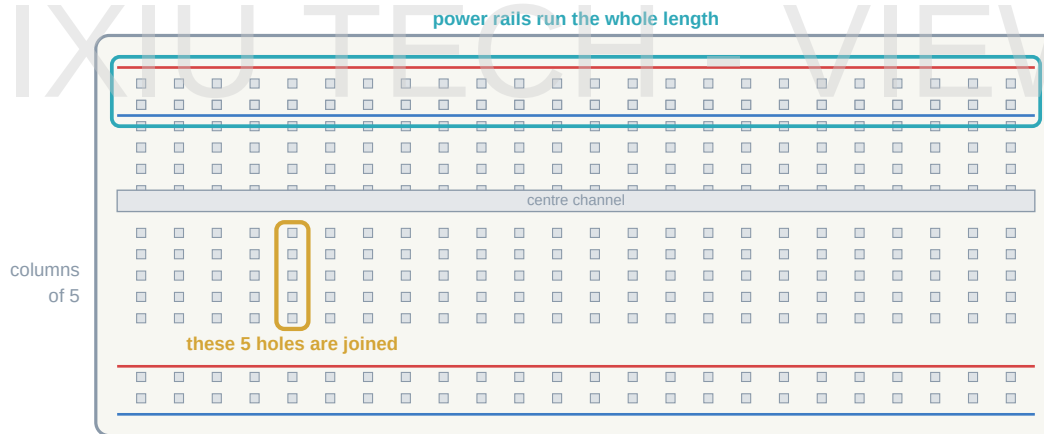


Figure 1.3 - Inside a breadboard: which holes are secretly connected

- The **long side rails** marked + and - run the whole length. Use them for power and ground.
- In the middle, holes are joined in **columns of five**, up and down only, never across.
- The **centre channel** splits the board in two. Nothing crosses it.
- So two legs in the same column of five are connected; two legs in different columns are not.

Try This

Push an LED into the breadboard with its legs in different columns. Put a 220 ohm resistor in the same column as the long leg. Wire the resistor's free end to 5V and the short leg's column to GND. Plug in the USB - it glows, with no code at all. Now pull out the GND wire and watch it die. You have just proved the loop rule with your own hands.

2.2 Two Kinds of Signal

A light switch has exactly two positions, on and off, with nothing in between. Engineers call that a **digital** signal. A fan regulator moves smoothly through many positions - that is an **analog** signal. The real world is mostly analog: temperature climbs, light fades, distance changes continuously.

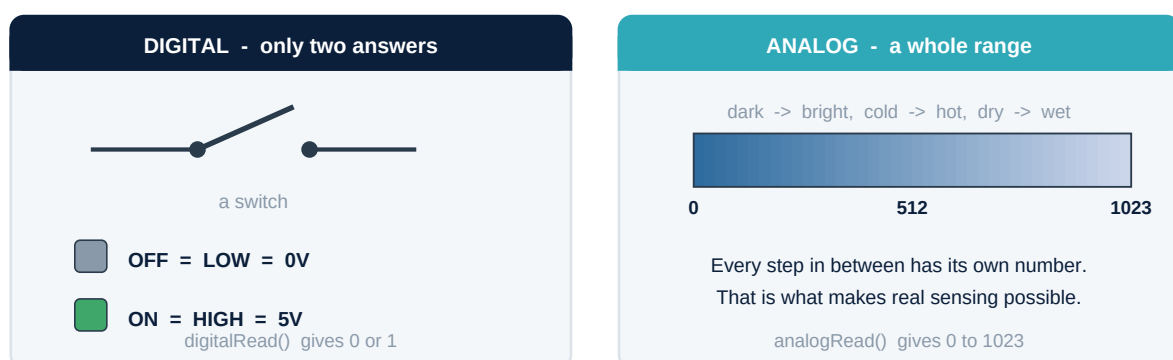


Figure 1.2 - The big idea of Class 7: numbers instead of yes-or-no

The Arduino handles both. Digital pins D0 to D13 deal in HIGH and LOW. Analog pins A0 to A5 measure a voltage and report a number from 0 to 1023. Here is the interesting part: the HC-SR04 you will use all year is a **digital** sensor, even though distance is an analog quantity. It does its own measuring inside and hands you the answer as the *length of a pulse*. Working out what that pulse means is your job in Unit 3.

New word	What it means
Voltage	How hard electricity is pushed. Measured in volts.
Resistance	Anything that makes current flow harder. Measured in ohms.
Ground (GND)	The return path. Every circuit needs one.
Digital / Analog	Only two values / any value within a range.